

Practice problems (Final term)

Z-transform

Example 3.2.2

Determine the z -transform of the signals

(a) $x(n) = (\cos \omega_0 n)u(n)$

(b) $x(n) = (\sin \omega_0 n)u(n)$

Example 3.2.4

Determine the transform of the signal

$$x(n) = \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{elsewhere} \end{cases} \quad (3.2.6)$$

Example 3.2.5

Determine the z -transforms of the signals

(a) $x(n) = a^n (\cos \omega_0 n)u(n)$

(b) $x(n) = a^n (\sin \omega_0 n)u(n)$

Example 3.2.6

Determine the z -transform of the signal

$$x(n) = u(-n)$$

For Digital Filters design

Practice the solved examples in class, relevant examples, and Assignment 3.